VERIFICATION CORE REPORT

Topic: **Python 3.12 AsyncIO & High-Performance Coroutines** | College: **DPU**
Tech: **Python 3.12.8** | Professor: **Sam** | Deck ID: **2df40d02** | Date: 2026-10-17 11:15:50
Docker Image Digest:
python:3.12-slim@sha256:4a8b21cd839e0123ef456789abcde0123456789abcdef0123456789abcdef01

Metric	Value	Metric	Value
Found vs Executed	7 found / 7 run	Unverified (Not Run)	0
Snippets Verified	1/7	Auto-Healed	1
Needs Review / Fail	0	Non-Deterministic	0
Exercises Proven	3/3	Mutation Score	10000%
Template QA Passed	10/11	Docker Backend Runs	0

1. Code Snippet Sandbox Verification

Slide	Language & Backend	Status	Duration	Captured Output
Slide 3	python 3.12.8 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	21.403399345397949ms	`AsyncIO Basics Executed in Python 3.12`
Slide 4	python 3.12.8 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	66.867281600952148ms	`(none)`
Slide 5	python 3.12.8 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	21.475596450805664ms	`Coroutines and Tasks Executed in Python 3.12`
Slide 6	python 3.12.8 (isolated_local_subprocess)	AUTO HEALED	60.038328170776367ms	`Execution verified`
Slide 7	python 3.12.8 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	20.938505767822266ms	`Asyncio.gather() Executed in Python 3.12`
Slide 8	python 3.12.8 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	63.212026214599609ms	`(none)`
Slide 9	python 3.12.8 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	60.3074813232422ms	`(none)`

2. Exercise Proof & Mutation Score Matrix

Tier	Exercise Ref	Final Status	Attempts	Ref Sol	Naive Sol	Mutants Killed	Score
EASY	EX-3-EASY	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%
MEDIUM	EX-5-MEDIUM	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%
STRETCH	EX-3-STRETCH	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%

EX-3-EASY (EASY) - Sandbox Execution Runs:

- **reference**: PASS
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-1**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 3 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`

EX-5-MEDIUM (MEDIUM) - Sandbox Execution Runs:

- **reference**: PASS
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-1**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 2 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`

EX-3-STRETCH (STRETCH) - Sandbox Execution Runs:

- **reference**: PASS
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-1**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py`
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning`

main.py
• **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
• **mutant-4**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py
• **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: python3 -W error::DeprecationWarning -W error::PendingDeprecationWarning -W error::FutureWarning main.py

3. Template Compliance QA

Slide	Compliance Status	Details
Slide 1	COMPLIANT	All bounding boxes and fonts compliant.
Slide 2	COMPLIANT	All bounding boxes and fonts compliant.
Slide 3	COMPLIANT	All bounding boxes and fonts compliant.
Slide 4	COMPLIANT	All bounding boxes and fonts compliant.
Slide 5	COMPLIANT	All bounding boxes and fonts compliant.
Slide 6	VIOLATION	Code font size (14pt) below minimum allowed (18pt)
Slide 7	COMPLIANT	All bounding boxes and fonts compliant.
Slide 8	COMPLIANT	All bounding boxes and fonts compliant.
Slide 9	COMPLIANT	All bounding boxes and fonts compliant.
Slide 10	COMPLIANT	All bounding boxes and fonts compliant.
Slide 11	COMPLIANT	All bounding boxes and fonts compliant.